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DE LA SALLE UNIVERSITY
College of Computer Studies
Department of Computer Technology

Integrative Programming
Machine Project Proposal & Specification

Group Name: Bonmardillsy Section: S15 & S16

Group Members:
1. Bondoc, Charles Andrew G
2. Marcelino, Adriane Joaquin C.
3. Padilla, Terrence M.
4. Sy, Justin Loui D.

Project Overview

1.1 Project Title

UniCanteen: Centralized Campus Food Ordering & Management System
PPT LINK: [Presentation Link]

1.2 Project Description

The project aims to streamline the purchasing process to help customers quickly check the availability and variety of products at the canteen. It does this by aggregating all of the food stalls and their products into an intuitive web application that customers can access before going to the canteen to order ahead of time.

Through this application, it helps solve the problem of stalls communicating with customers regarding their stocks, customers navigating the different food options, as well as simplifying the queuing process commonly found in canteens by being able to order food in advance and avoiding long lines.

1.3 Target Users

The system serves the following demographic:

- *Students*
- *Faculty*
- *Staff*
- *Food Vendors/Stalls (Bloemen Hall Stalls, Agno Food Court, Kitchen in St. La Salle Building etc.)*

Proposed System Features

Part 1: Client Side

The client-side of the system is designed for students, faculty and staff who will use the browser for food options and place orders within the canteens. Without administrative access, this focuses to provide an easy to use interface that allows users to view available and upcoming food stalls and their corresponding products. Through the web application, it allows the users or customers to place food or payment in advance, making the buying experience much easier and more convenient. The following features of the client-side system are:

1. Menu Browsing and Product Selection
    - Users can browse through a consolidated list of food stalls and view menu items offered by each stall. Each product listing includes relevant details such as product name, price and availability status, enabling the user to make their purchases before going to the canteen and stalls.

2. Order Checkout and Tracking
    - Users may choose available payment options by the system during the checkout process. The food stalls will have a reference for order details and selected payment method. After, the user can monitor the status of their order whether it is pending, being prepared or ready to pick up..

3. Ratings and Comment
    - Users may rate the food by giving 1 to 5 stars during the post-checkout process. They may also give any comments or suggestions to improve the products or the services. The average ratings stars of the stall/food will be displayed in the menu sections.

The limitations of the client-side include no offline connectivity as being a web application, it requires an internet connection to work. Additionally, the mobile payment app (G-Cash) and card payment will be payment options only. Furthermore, no order cancellations are allowed meaning that once a customer's food has been ordered, it cannot be canceled to prevent food waste and financial losses for the food stall. Lastly, payments will be done and dealt with by the food stall itself.

Part 2: Server Side

A. Roles

The server-side system serves as the central management system for administrative users, including campus management (System Admin) and individual food stall owners (vendor Admin). The server-side system application is secured via login authentication and is responsible for managing data that would populate the client-side of the system.

A. Food Vendors/Stalls (Vendor Admin)
   - Vendor admins or stall managers operating within the canteen. Their role focuses on operational efficiency and inventory management.
   - Inventory Controller: Responsible for keeping the digital menu synchronized with the physical kitchen by toggling item availability
   - Order Manager:Tasked with monitoring the incoming queue in real-time and updating order statuses to keep customers informed.
   - Sales Reviewer:Uses the system's reporting tools to track daily revenue and identify best-selling items to adjust future food preparation.

B. System Administrator (Campus Management)
   - Campus management responsible for creating vendor accounts and overseeing user compliance.

B. Features

The following features of the server-side system are:

1. Vendor & Food Menu Management
   - Vendor Admins would be provided with Create, Read, Update, and Delete capabilities to manage their menu items. This allows them to modify menu items, such as changing food images, setting detailed descriptions, and updating prices to match their physical offerings, ensuring flexibility in that they are able to add/remove/change items around freely. They would be able to toggle or mark items as "Available" or "Sold out.". This status is committed immediately to the database and is reflected on the client-side server to prevent users from ordering out-of-stock items.

2. Orders Dashboard
   - Food stalls would be able to see their incoming order queue, which displays student orders in real-time. It helps vendors manage the workflow by updating order statuses through a defined cycle, moving from Preparing to Ready for Pickup, and Completed. Moreover, adding physical

transactions is another feature to ensure data consistency with inventory and sales. These state changes are synchronized with the database to ensure customers receive accurate information on their side.

3. Sales Monitoring
   - To assist with business operations, the system would include a monitoring tool that presents a view of all successfully fulfilled orders. Since payments are collected physically, this module relies on the vendor marking orders as "Completed" to generate a history of transactions. Vendors can filter these by date or status and view basic performance summaries, like Total Revenue and Most Selling Menu Items. This would help them track daily earnings and adjust their inventory, referring to each sold meal, based on what's popular and in demand.

4. System Administration
   - The System Admin role would focus on system oversight, responsible for implementing new food stalls by creating vendor accounts and assigning secure credentials. Furthermore, this module provides capabilities for user oversight, allowing the administrator to view registered student and staff accounts and manage access privileges. This also gives them the ability to ban users who violate the platform policies, such as placing fake orders.
5. Queuing
   - Upon cashless payment, a queuing number will be displayed which should be brought to the stalls for order verification. Once an order has been verified and picked up, the vendor will update the state to “Completed” to finalize and record the sale.

The server-side system has its limitations, primarily the absence of an integrated online payment gateway. The system functions on a "QR/Cashless" basis, where payment is done as the order is processed to lessen the financial risk. Additionally, the system is designed strictly for web browsers and does not support native mobile push notifications, meaning vendors must maintain an active dashboard session to monitor incoming orders.

Tab 2

Integrative Programming
Machine Project Proposal & Specification

Introduction

This machine project serves as a culminating requirement for the Integrative Programming course. It is designed to assess students' ability to integrate multiple web development technologies—such as HTML for structure, CSS for presentation, PHP for server-side logic, and databases for data management—into a cohesive and functional web-based system. The project emphasizes proper coding practices, separation of concerns, and the practical application of programming concepts discussed throughout the course. Students are expected to demonstrate not only technical proficiency but also the ability to design systems that address real-world problems in an organized and systematic manner.

The system to be developed must be aligned with the operational needs of a specific organization or institution, whether real or hypothetical. Each project group is required to identify an organizational context (e.g., school, small business, clinic, retail store, government office, or service-oriented organization) and **clearly define the problem or inefficiency that the proposed web-based system aims to address.**

Although having an actual client is not required, the proposed solution must be realistic, relevant, and grounded in real-world scenarios. The design and functionality of the system should demonstrate how it can effectively support organizational processes, improve efficiency, enhance data management, or provide better services to its intended users.

The client-side system should focus on meeting the needs of end users within the organization, while the server-side (admin) system should support administrative tasks such as data management, monitoring, reporting, and system configuration. Together, both systems must reflect a coherent solution to the identified organizational problem.

Project Proposal Submission and Presentation

Students must use the provided template to clearly describe and define the specifications of their proposed project.

- This group project is composed of four (4) members only.
- The tentative Project Proposal Presentation will be held on Week 4.
- During the proposal presentation, the project scope and features will be discussed with the instructor.
- After the presentation, groups are required to revise and resubmit their

proposal.
- The resubmitted proposal (due one week after the presentation) will be considered the final and approved project specification.

Development Tools and Technologies

- HTML & CSS – for structure and styling
- PHP – for server-side logic
- MySQL – for database management
- JSON and XML may be used for data exchange and integration
- JavaScript (Front-end only)
    - UI behavior and interactivity
    - DOM manipulation
    - Form validation
    - AJAX-based data fetching

Important: The use of JavaScript for server-side development (e.g., Node.js, Express, backend APIs, or business logic) is strictly prohibited.

Integration Tasks

- The client-side and server-side systems must be properly integrated using any appropriate method (e.g., data integration or application-level integration).
- Data between the two systems must be consistent, synchronized, and updated correctly. Any changes made through the admin system should be reflected accurately on the client-side system.

Final Project Submission Guidelines

Each group must submit the following:

- Complete Source Code
- Entity Relationship Diagram (ERD)
- Exported Database File (sql)

Grading and Presentation Guidelines

- The Final Machine Project Presentation will be held on Week 14.
- Each group will be given 30 minutes, inclusive of:
    - System demonstration
    - Explanation of features
    - Question and Answer portion
- The project will be evaluated based on the approved proposal. Any feature not included in the final proposal may not be considered during grading.
- Grading will follow the official rubric provided by the instructor.